

SOC ANALYST PORTFOLIO CASE STUDY

Simulated Web Attack Investigation: Packet Capture and Incident Response

Wireshark traffic analysis, Apache and SSH log correlation, incident documentation, and evidence integrity

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OUTCOME Captured and correlated an authorized sequence of network reconnaissance, two web-attack simulations, and three failed SSH logins. The server rejected every suspicious action; no compromise or service outage occurred.

Component	Address / role	Control
Parrot Security 7.3	10.10.10.5 - simulation workstation	Wireshark 4.4.16, Nmap 7.95, curl, SSH
Ubuntu Server 24.04.4 LTS	10.10.10.4 - authorized target	Apache 2.4.58, OpenSSH, host logs
SOC-Lab	10.10.10.0/24 isolated NAT network	No public or production target

Portfolio note: Authorized local simulation only. Wazuh was not required for this exercise; the investigation emphasizes packet capture, endpoint logs, cross-source correlation, and stakeholder communication. AI assistance is disclosed in this report.

01 / EXECUTIVE SUMMARY

What happened and what it means

On 24 August 2026, controlled activity was generated from a Parrot Security workstation against one Ubuntu web server inside the isolated SOC-Lab. Wireshark captured the traffic while Apache and OpenSSH recorded application and authentication events.

FINAL ASSESSMENT No successful exploitation, login, data access, persistence, or availability impact was observed.

Objectives

- Capture a clean baseline before suspicious activity.
- Identify reconnaissance patterns at the packet level.
- Observe path traversal and SQL-injection-shaped HTTP requests.
- Correlate three failed SSH attempts with packet and host logs.
- Preserve the PCAP, logs, commands, timestamps, and hashes.

Scope and controls

- Single authorized target: 10.10.10.4.
- Source restricted to 10.10.10.5.
- Low-rate Nmap timing (-T3); three manual SSH attempts only.
- No brute-force tooling, exploit execution, or production access.
- Services remained available throughout the test.

Signal	Observed result	Interpretation
Reconnaissance	22/tcp and 80/tcp open; 8080/tcp filtered	Limited service discovery
Path traversal	HTTP 400 Bad Request	Request rejected
SQL injection string	HTTP 404 Not Found	Endpoint absent / no query execution shown
SSH password attempts	Three denials for invalid user	No authentication success

02 / INVESTIGATION WORKFLOW

Repeatable SOC workflow

- 01 Scope and validate**
Confirm host identities, IP addresses, services, and authorization boundary.
- 02 Create evidence structure**
Create pcap, logs, and notes folders; record a start timestamp.
- 03 Start filtered capture**
Capture only traffic involving 10.10.10.4 on Parrot interface enp0s3.
- 04 Establish baseline**
Send a labeled HTTP GET and confirm the expected 200 response.
- 05 Generate reconnaissance**
Run a low-rate SYN scan against ports 22, 80, and 8080.
- 06 Generate web activity**
Send labeled path traversal and SQL-injection-shaped requests.
- 07 Generate authentication failures**
Attempt three manual SSH logins to a nonexistent account.
- 08 Correlate evidence**
Compare Wireshark, Apache access.log, and auth.log timestamps and fields.
- 09 Preserve and verify**
Save the PCAP, export log excerpts, record end time, and hash six artifacts.

KEY COMMANDS AND CONTROLS

```
wireshark -i enp0s3 -f "host 10.10.10.4"
nmap -sS -T3 -p 22,80,8080 --reason 10.10.10.4
curl --path-as-is -A "SOC-Lab-Path-Traversal" ...
curl --path-as-is -A "SOC-Lab-SQLi-Test" ...
ssh -o PubkeyAuthentication=no labintruder@10.10.10.4
sha256sum pcap/*.pcapng logs/*.log notes/*.txt
```

03 / BASELINE AND RECONNAISSANCE

A known-good request established context

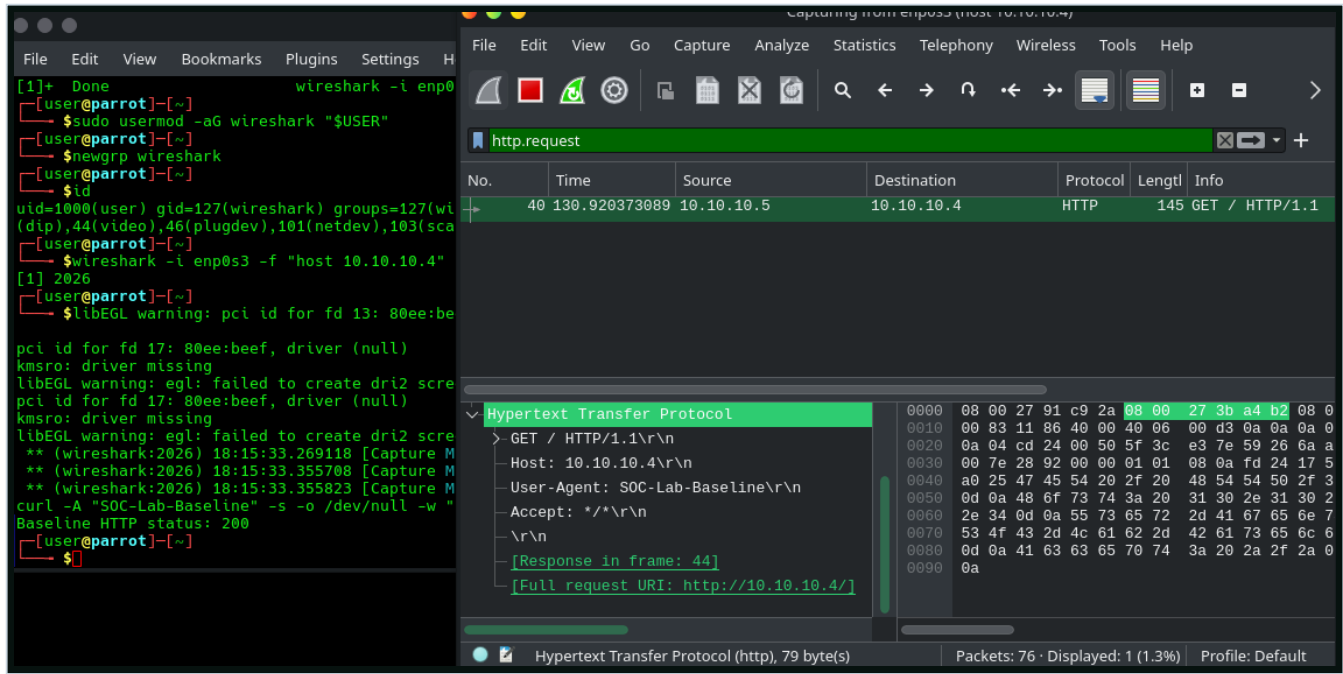


Figure 1. Wireshark baseline: HTTP GET / from 10.10.10.5 to 10.10.10.4, labeled with User-Agent SOC-Lab-Baseline. Apache returned HTTP 200.

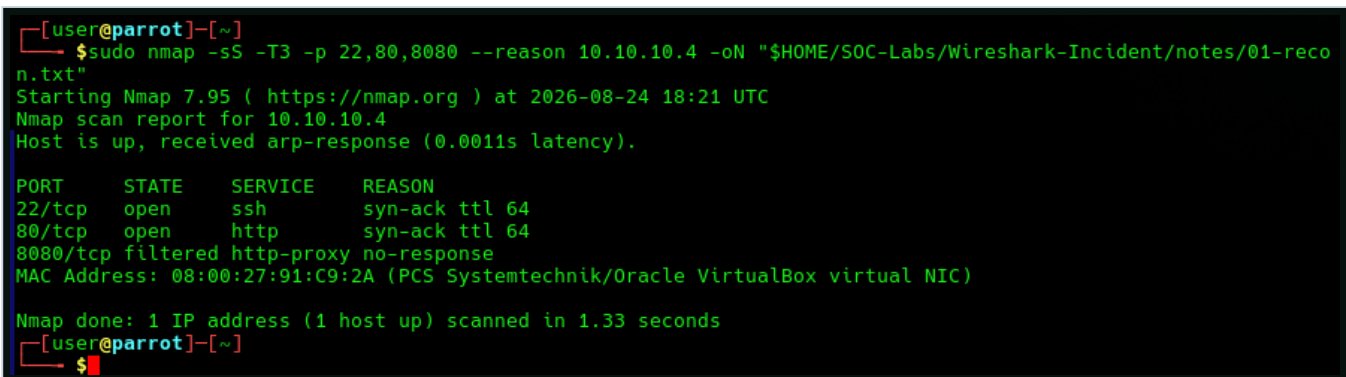


Figure 2. Controlled SYN reconnaissance identified SSH (22) and HTTP (80) as open and TCP/8080 as filtered with no response.

ANALYST INTERPRETATION The baseline separated expected service traffic from later suspicious requests. The scan revealed exposure, not compromise.

04 / PACKET ANALYSIS

Reconnaissance pattern visible in TCP flags

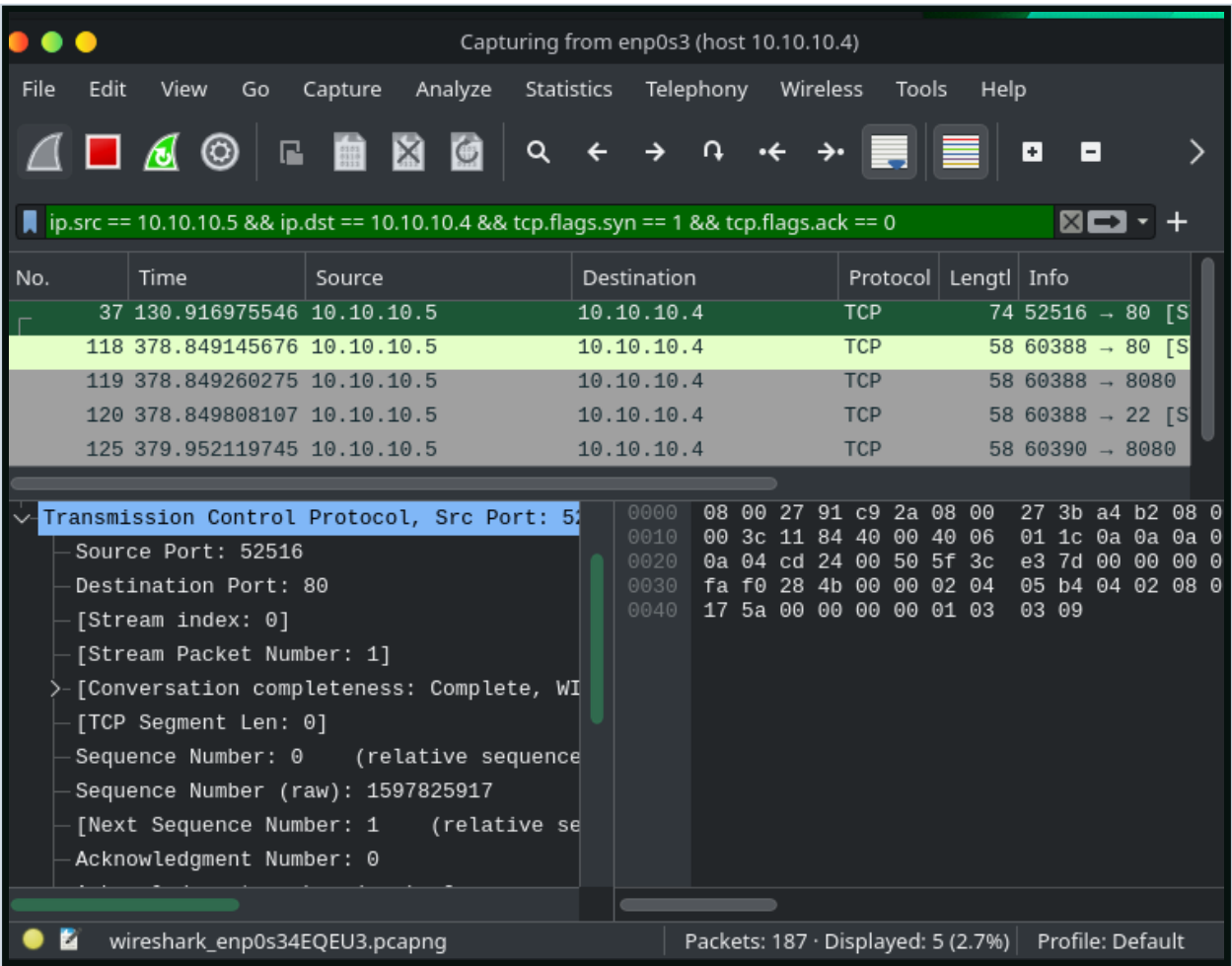


Figure 3. Wireshark display filter isolates initial SYN packets from 10.10.10.5 to 10.10.10.4. The destination sequence covers 80, 8080, and 22, matching the saved Nmap command.

Field	Evidence	Meaning
Source	10.10.10.5	Authorized Parrot workstation
Destination	10.10.10.4	Ubuntu web server
Flags	SYN=1, ACK=0	Initial connection probes
Response	SYN-ACK on 22 and 80	Services accepted connections
8080	No response / filtered	Traffic was filtered or silently dropped

05 / WEB ATTACK SIMULATION

Path traversal request was rejected

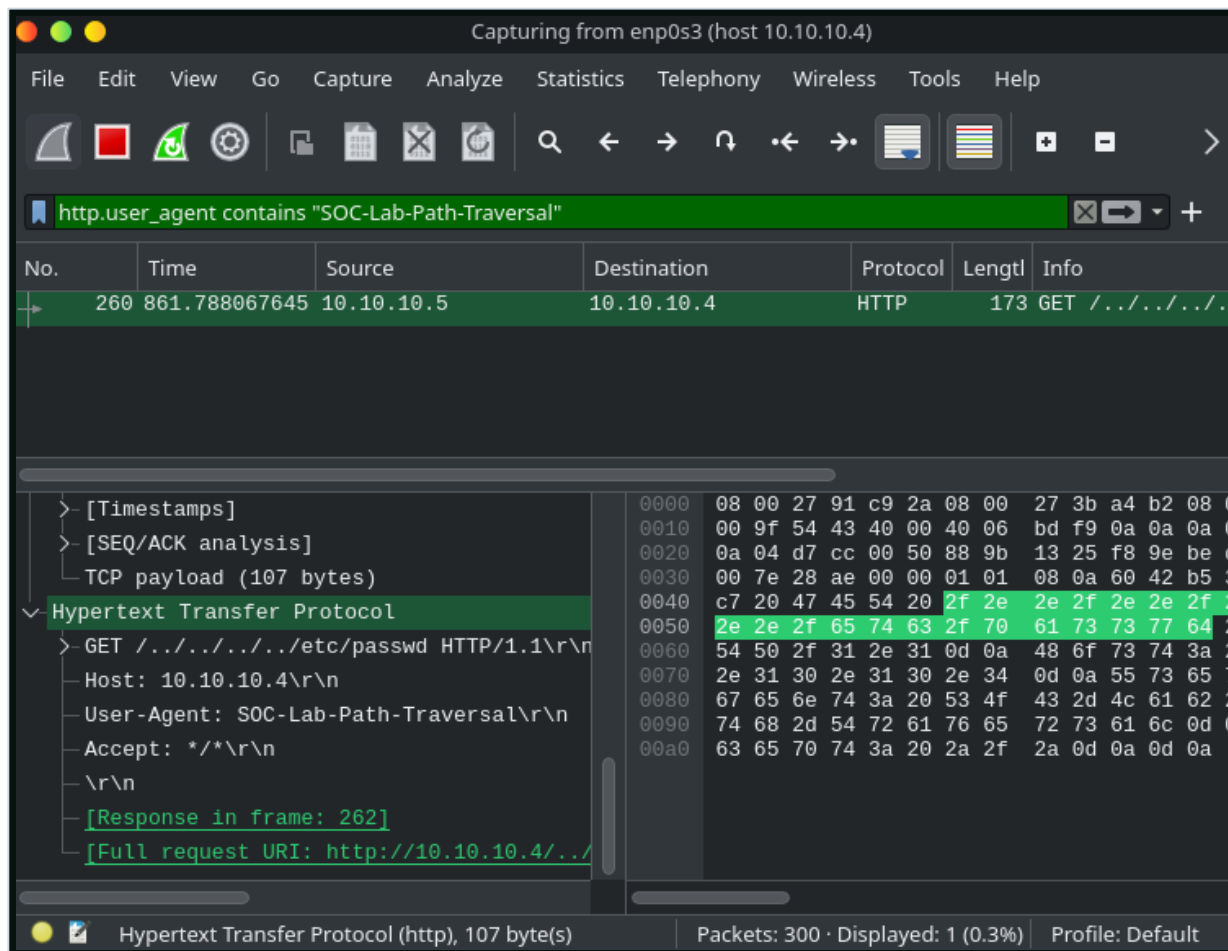


Figure 4. Wireshark expands the HTTP request: GET ../../../../../../etc/passwd with the unique User-Agent SOC-Lab-Path-Traversal.

FINDING HTTP 400 indicates that Apache rejected the malformed request. The packet proves the attempt reached the server; it does not show that /etc/passwd was disclosed.

Evidence chain The source terminal recorded HTTP 400; the packet contains the exact URI and label; Apache access.log recorded the same URI, source IP, timestamp, label, and status code.

06 / WEB ATTACK SIMULATION

SQL-injection-shaped request reached an absent endpoint

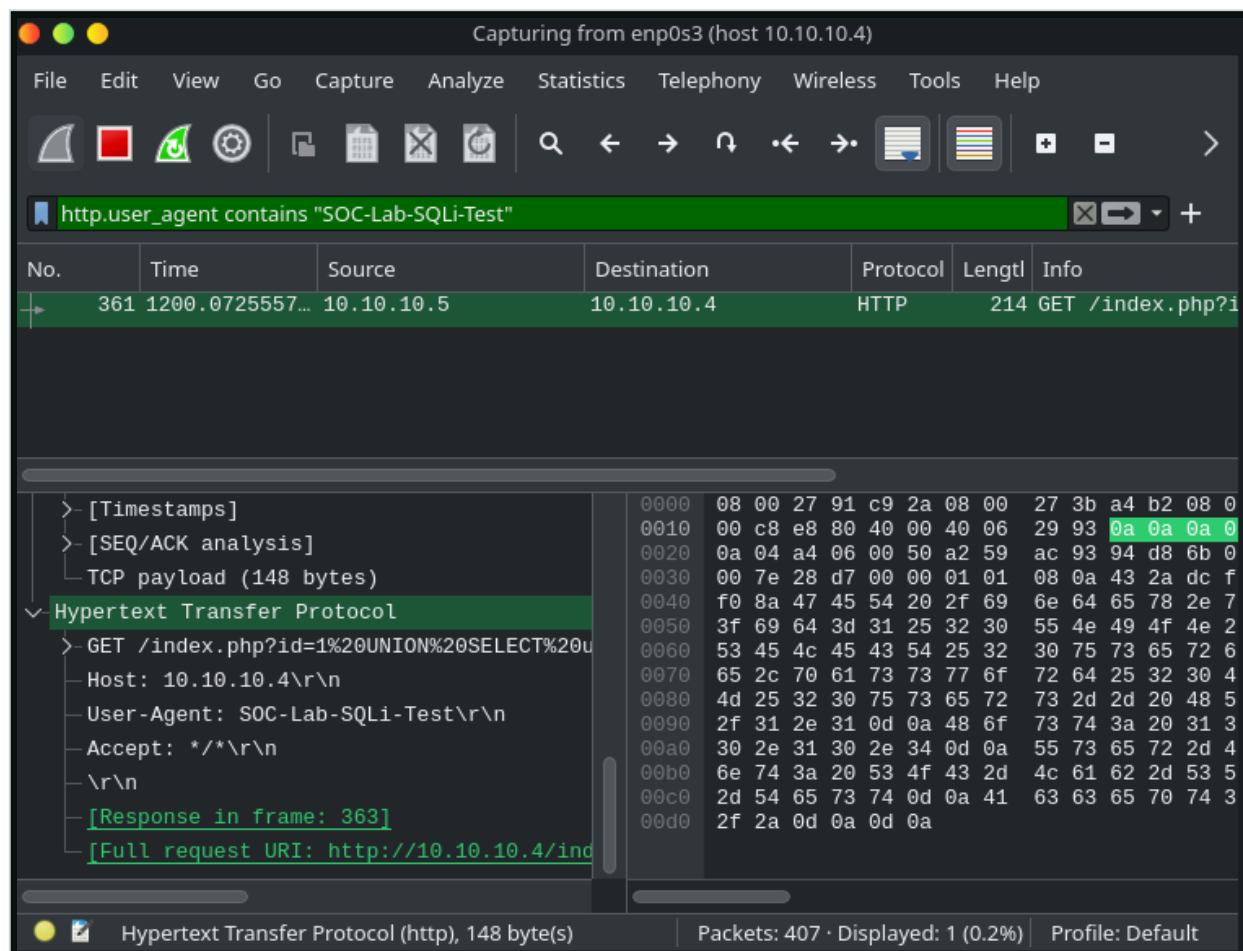


Figure 5. Wireshark shows the encoded UNION SELECT payload and the label SOC-Lab-SQLi-Test inside the HTTP request.

FINDING HTTP 404 means the requested index.php resource was not present. The payload was observed in transit and logged, but the evidence does not show database execution or data exposure.

Analytical discipline A suspicious string alone is not proof of exploitation. The correct classification is attempted application-layer probing with no demonstrated impact.

07 / AUTHENTICATION ACTIVITY

Three failed SSH attempts produced host evidence

```
socstudent@linux-webserver01:~$ sudo ss -lntp | grep -E ':(22|80)[[:space:]]'
LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*      users:((("sshd",pid=881,fd=3),("systemd",pid=1,fd=93))
LISTEN 0      4096      [::]:22        [::]:*        users:((("sshd",pid=881,fd=4),("systemd",pid=1,fd=94))
LISTEN 0      511      *:80          *:~          users:((("apache2",pid=887,fd=4),("apache2",pid=886,fd=4),("apache2",pid=885,fd=4))
socstudent@linux-webserver01:~$ sudo grep -E 'labintruder|Failed password|Invalid user' /var/log/auth.log | tail -n 12
[sudo] password for socstudent:
2026-08-24T18:44:36.442051+00:00 linux-webserver01 sshd[2240]: Invalid user labintruder from 10.10.10.5 port 53134
2026-08-24T18:44:51.120637+00:00 linux-webserver01 sshd[2240]: Failed password for invalid user labintruder from 10.10.10.5 port 53134 ssh2
2026-08-24T18:44:51.401599+00:00 linux-webserver01 sshd[2240]: Connection closed by invalid user labintruder 10.10.10.5 port 53134 [preauth]
2026-08-24T18:46:00.788589+00:00 linux-webserver01 sshd[2246]: Invalid user labintruder from 10.10.10.5 port 39054
2026-08-24T18:46:12.362714+00:00 linux-webserver01 sshd[2246]: Failed password for invalid user labintruder from 10.10.10.5 port 39054 ssh2
2026-08-24T18:46:13.459768+00:00 linux-webserver01 sshd[2246]: Connection closed by invalid user labintruder 10.10.10.5 port 39054 [preauth]
2026-08-24T18:46:22.816637+00:00 linux-webserver01 sshd[2248]: Invalid user labintruder from 10.10.10.5 port 45842
2026-08-24T18:46:32.464335+00:00 linux-webserver01 sshd[2248]: Failed password for invalid user labintruder from 10.10.10.5 port 45842 ssh2
2026-08-24T18:46:33.482703+00:00 linux-webserver01 sshd[2248]: Connection closed by invalid user labintruder 10.10.10.5 port 45842 [preauth]
2026-08-24T18:48:34.605350+00:00 linux-webserver01 sudo: socstudent : TTY=ttty1 ; PWD=/home/socstudent ; USER=root ; COMMAND=/usr/bin/grep -E 'labintruder|Failed password|Invalid user' /var/log/auth.log
socstudent@linux-webserver01:~$
```

Figure 6. Ubuntu auth.log records invalid user labintruder, failed password, and pre-authentication disconnect events from 10.10.10.5 across three source ports.

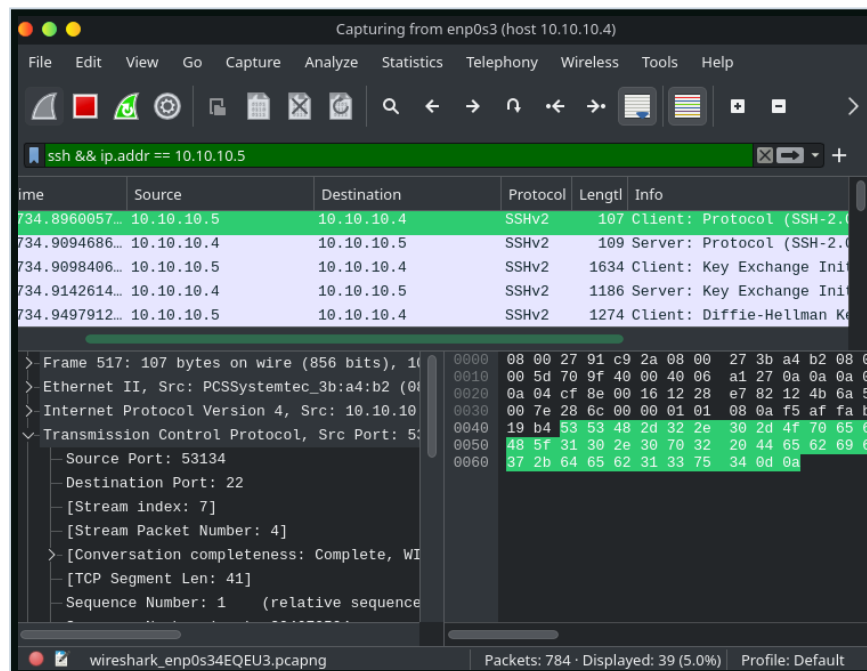


Figure 7. Wireshark isolates SSHv2 negotiation. Source port 53134 corresponds to the first auth.log attempt, linking network and endpoint evidence.

FINDING All attempts were denied. No valid account, successful session, or post-authentication activity was observed.

08 / MULTI-SOURCE CORRELATION

One sequence, three independent views

```
socstudent@linux-webserver01:~$ sudo grep 'SOC-Lab-' /var/log/apache2/access.log | tail -n 10
10.10.10.5 - - [24/Aug/2026:18:17:47 +0000] "GET / HTTP/1.1" 200 11148 "-" "SOC-Lab-Baseline"
10.10.10.5 - - [24/Aug/2026:18:29:58 +0000] "GET /../../../../etc/passwd HTTP/1.1" 400 656 "-" "SOC-Lab-Path-Traversal"
10.10.10.5 - - [24/Aug/2026:18:35:36 +0000] "GET /index.php?id=1%20UNION%20SELECT%20username,password%20FROM%20users-- HTTP/1.1" 404 607 "-" "SOC-Lab-SQLi-Test"
socstudent@linux-webserver01:~$
```

Figure 8. Apache access.log preserves the labeled baseline, path traversal, and SQLi-shaped requests with timestamps and HTTP statuses.

Time (UTC)	Activity	Network evidence	Host evidence
18:17:47	Baseline GET /	HTTP request captured	Apache 200
18:21	Port reconnaissance	SYNs to 22/80/8080	22 and 80 listening
18:29:58	Path traversal	URI and User-Agent captured	Apache 400
18:35:36	SQLi-shaped request	Encoded payload captured	Apache 404
18:44–18:46	Three SSH attempts	SSHv2 sessions captured	Invalid user / denied

Confidence statement: Source and destination IPs, request labels, URIs, HTTP status codes, timestamps, and SSH source ports align across the packet capture and server logs. This cross-source agreement supports a high-confidence reconstruction.

SOC PRACTICE No single artifact carries the conclusion. Confidence comes from corroboration between network telemetry and endpoint records.

09 / EVIDENCE PRESERVATION

Integrity manifest for six artifacts

```
[user@parrot]~/SOC-Labs/Wireshark-Incident
$sha256sum pcap/*.pcapng logs/*.log notes/01-recon.txt notes/lab-*.txt | tee notes/evidence-sha256.txt
e137e22b21f5358def0b4ae02d704805d452093c81131d921072a00f0ee744f7 pcap/web-attack-incident-2026-08-24.pcapng
adc4161e50b8e453f67776d3153e66e3ce934f06116e31a0e524c217fcff28b7 logs/apache-incident-evidence.log
52f2712c13191c20cea9a2e282b27f85fda1c85b6a0f59afd92d265e47065f45 logs/ssh-incident-evidence.log
0a81414132df9bf3f557e7310f32e1a299626a6de65c770670f0b785a78f3149 notes/01-recon.txt
9e5264a87a94d177545db0987369995655be990dc026a34096e8527a2b2df54c notes/lab-end.txt
cefc9d6f950c6e7d60dbededd294bcfcb3721d7716e1e01f8f34d6e2fc9aaf14 notes/lab-start.txt
[user@parrot]~/SOC-Labs/Wireshark-Incident
$
```

Figure 9. Final SHA-256 manifest covers the packet capture, Apache and SSH excerpts, reconnaissance output, and start/end timestamps.

Artifact	SHA-256 (abbreviated)
web-attack-incident-2026-08-24.pcapng	e137e22b...e744f7
apache-incident-evidence.log	adc4161e...ff28b7
ssh-incident-evidence.log	52f2712c...065f45
01-recon.txt	0a814141...8f3149
lab-start.txt	cefc9d6f...9aaf14
lab-end.txt	9e5264a8...2df54c

Why this matters A hash allows a later reviewer to verify that an evidence file has not changed since preservation. The saved PCAP contained 784 packets; display filters were used for analysis without altering the underlying capture.

EVIDENCE QUALITY Evidence handling was reproducible and auditable: named artifacts, bounded timestamps, copied host excerpts, and cryptographic integrity checks.

10 / FINDINGS AND RECOMMENDATIONS

Attempted activity without demonstrated impact

Finding	Observed severity	Rationale
Service reconnaissance	Low	Scoped, low-rate discovery; no exploitation
Path traversal attempt	Moderate signal	Malformed request rejected with HTTP 400
SQLi-shaped request	Moderate signal	Endpoint absent; HTTP 404; no database evidence
SSH password guessing	Moderate signal	Three denials; no valid user or session
Overall incident impact	None observed	No access, disclosure, persistence, or outage

Recommended controls

- Minimize exposed services and keep TCP/8080 filtered unless explicitly required.
- Use SSH keys, disable direct password authentication where operationally appropriate, and consider rate limiting or fail2ban.
- Maintain Apache and OpenSSH patching; review application input validation and routing before exposing dynamic endpoints.
- Alert on port-scan patterns, traversal strings, SQLi signatures, repeated invalid users, and authentication bursts.
- Preserve packet captures and log excerpts with synchronized time and SHA-256 hashes during investigations.

MITRE ATT&CK context

Technique	Context in this exercise
T1046 Network Service Discovery	Port enumeration of 22, 80, and 8080
T1190 Exploit Public-Facing Application	Context for traversal/SQLi-shaped attempts; target was local, not public-facing
T1110.001 Password Guessing	Three manual password attempts against an invalid SSH user

11 / PORTFOLIO VALUE AND TRANSPARENCY

Skills demonstrated

Capability	Evidence in this case study
Packet analysis	Capture filters, display filters, TCP flags, HTTP fields, SSHv2 sessions
Log analysis	Apache access.log and OpenSSH auth.log interpretation
Incident reconstruction	Timeline and cross-source correlation
Security assessment	Impact distinction: attempt versus successful exploitation
Evidence handling	Named artifacts, timestamp bounds, SCP transfer, SHA-256 manifest
Communication	Technical report plus business-focused stakeholder recap

RESUME-READY PROJECT SUMMARY Investigated a controlled multi-stage web and SSH event using Wireshark, Nmap, Apache, and OpenSSH logs; correlated packet and endpoint evidence, confirmed no compromise, and preserved six hashed artifacts.

AI assistance and authorship

OpenAI’s ChatGPT (called “Sarah” by the author) assisted with troubleshooting guidance, explanatory context, document organization, editing, and source identification. Ryan Bryant defined and authorized the lab scope, executed every command, captured the screenshots and logs, reviewed the evidence, and made the final assessment. The report does not claim that AI performed the hands-on work.

Official validation sources

Wireshark User’s Guide • Nmap SYN Scan • Nmap Port States
Apache HTTP Server Log Files • MITRE T1046 • MITRE T1190 • MITRE T1110.001

FINAL DISPOSITION Controlled simulation completed. Evidence supports reconnaissance and attempted access patterns, but no successful exploitation or unauthorized access.